

Zaid Duraid

zaidduraid.com | linkedin.com/in/zaid-duraid | github.com/Zaid-Duraid

EDUCATION

Memorial University of Newfoundland

St. John's, NL, Canada

Bachelor of Engineering: **Computer Engineering** - CGPA: 4.00/4.00 (95.3%)

Sep. 2022 - Apr. 2027

- **Dean's List (2022-2025)**
- **Scholarships:** Memorial University Alumni Entrance, Innovasea Computer Engineering, Verafin Inc. Computer Engineering, Charlie Sheppard Memorial - Hatch, PEGNL Past-President, Bob Thorburn Memorial, Gander Lions Club Community Leaders, NLESD Education Foundation, Town of Gander

EXPERIENCE

Drive Inverter (DI) Firmware Validation Engineer Intern

Jan. 2026 - Present

Tesla Inc.

Palo Alto, California, USA

- Implemented **LIN in Verilog on a Xilinx FPGA** for Tesla's processor-in-the-loop (PIL) validation, with fault injection and an AXI stream Linux driver, helping unblock PIL coverage for ~92% of DI firmware configurations
- Integrated a **MATLAB PMSR motor model onto a Xilinx FPGA in Verilog**, enabling real-time PIL dyno testing for DI firmware configurations not ready for high-load HIL, all within minutes of iteration time
- Developed **fault-handling validation tests** for verifying correct and timely fault responses across DI firmware configurations, including 6+ fault scenarios across gate driver, power, and electrical oil pump systems
- Drove adoption of **PIL developer and coding agent workflow improvements** such as optimized markdown documentation and embedded runtime-readable git-hash metadata in FPGA bitstreams

Advanced Research Electrical Engineer Intern

May 2025 - Aug. 2025

Solace Power

St. John's, NL, Canada

- Improved the design of a multi-MHz synchronous rectifier by adding a **PLL for closed-loop control** to maintain operation during ~100 us loss of switch gate driver signal
- Designed and built single-layer PCBs to quickly verify **integration of subcircuits** (e.g., RC inverter delay line for phase shift on GaN FET driver signal) into multilayer PCBs
- Researched **EMC solutions for a tens of MHz wireless power system**, reducing noise on key harmonics
- Wrote **C firmware for a TI F28003x microcontroller** to generate internal-clock or interrupt-driven PWM signals up to 9 MHz, with adjustable frequency, duty cycle, and phase shift via CANBUS

FPGA Hardware Engineer Intern

Jan. 2024 - Apr. 2024 and Sep. 2024 - Dec. 2024

Avalon Holographics Inc.

St. John's, NL, Canada

- Led the development of new **internal verification tools using Python** that integrated with existing testbenches to verify hardware image correction features for the holographic display and reduce iteration time by 10x
- Developed and documented **simulation testbenches for FPGA designs in SystemVerilog** that integrate with internal verification tools, reducing critical bug discovery from ~2 months on hardware to ~2-3 days in simulation
- Followed internal test processes by **assembling hardware, executing tests, and running TCL scripts** for the bring-up of new/refurbished PCBs and validation of FPGA bitstream releases

PROJECTS

Eastern Edge Robotics Student Design Team Software Co-Lead - Python, C/C++ Aug. 2023 - Dec. 2025

- Built **ROS2 applications on a Raspberry Pi** to interface with ROV hardware over UART and I2C
- Contributed to **two frontend piloting applications for browser-based and low-latency control**
- Implemented a **Gazebo ROV simulation environment** with C++ plugins for validating the full software stack
- Competed and presented alongside the team at the International MATE ROV Competition Explorer Class in 2023, 2024, and 2025, **placing in 11th, 9th, and 3rd place in the world out of around 20-30 teams**, respectively

Photosphere Board - KiCad

Dec. 2024 - Jan. 2025

- Designed a **PCB in KiCad** that interfaces with the Raspberry Pi Compute Module 4 and includes **dual MIPI CSI camera ports and USB 2.0** for multi-view video streaming in a space-constrained 5x10x5 cm environment

SKILLS

Software/Firmware/HDL: Verilog/SystemVerilog, C/C++. Python, Golang, Rust, Typescript

Hardware: Vivado, Quartus, Questa, KiCad, LTSpice, Altium

Tools: Git, Claude Code, Docker, ROS, Gazebo, Solidworks, Onshape